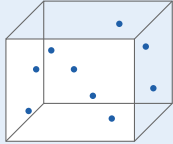


domain randomisation



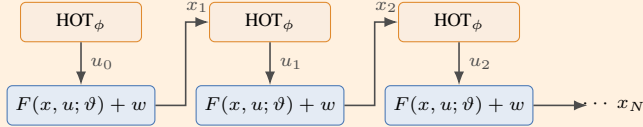
gusts w
sensor noise
initial trims ρ_0
targets ρ_t
both directions

ϑ, w

parameter box Θ

$C_{L0}, C_{L\alpha}, C_{D0}, k, \kappa, g_\theta, \lambda_i, \beta_i$
 $\vartheta \sim \mathcal{U}(\Theta)$ per episode

differentiable rollout of the identified model



gradients through the RK4 step of F and through the policy;
horizon curriculum $N\Delta$: 2 s $\rightarrow$ 20 s

LSM rows as a soft barrier while training, exact projection at run time

loss $\mathcal{J}(\phi)$

time $\sum_j \sigma(|V_j - \rho_t| - 0.3)$
energy $\sum_j (T_{r,j} + T_{c,j} V_j) \Delta$
barrier $\sum_{j,i} \text{softplus}(-h_i(x_j)/\epsilon)$
plant head $\|\hat{v}_j - v\|_{\Sigma-1}^2$
smoothness $\|u_j - u_{j-1}\|_R^2$

backpropagation through time, Adam: $\phi \leftarrow \phi - \eta_{lr} \nabla_\phi \mathcal{J}$